

Singer difference sets and the projective norm graph

Tamás Mészáros^{*†} Lajos Rónyai^{‡§} Tibor Szabó^{*¶}

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Abstract

We demonstrate a close connection between the classic planar Singer difference sets and certain norm equation systems arising from projective norm graphs. This, on the one hand leads to a novel description of planar Singer difference sets as a subset $\mathcal{H}$ of $\mathcal{N}$, the group of elements of norm 1 in the field extension $\mathbb{F}_{q^3}/\mathbb{F}_q$. $\mathcal{H}$ is given as the solution set of a simple polynomial equation, and we obtain an explicit formula expressing each non-identity element of $\mathcal{N}$ as a product $B \cdot C^{-1}$ with $B, C \in \mathcal{H}$. The description and the definitions naturally carry over to the nonplanar and the infinite setting. On the other hand, relying heavily on the difference set properties, we also complete the proof that the projective norm graph $\text{NG}(q, 4)$ does contain the complete bipartite graph $K_{4,6}$ for every prime power $q \geq 5$. This complements the property, known for more than two decades, that projective norm graphs do *not* contain $K_{4,7}$ (and hence provide tight lower bounds for the Turán number $\text{ex}(n, K_{4,7})$).

1 Introduction and results

1.1 Projective norm graphs

Let $\mathbb{F}$ be an arbitrary field, for $t \geq 2$ let $\mathbb{K}$ be a cyclic Galois extension of degree $t - 1$ and let us denote by N the norm of this extension, i.e. for $A \in \mathbb{K}$ we have $N(A) = A \cdot \phi(A) \cdot \phi^{(2)}(A) \cdots \phi^{(t-2)}(A)$ where the automorphism ϕ generates the Galois group of $\mathbb{K}/\mathbb{F}$ and $\phi^{(j)}$ denotes the j -fold iteration of ϕ . Then the *projective norm-graph* $\text{NG}(\mathbb{F}, \mathbb{K})$ has vertex set $\mathbb{K} \times \mathbb{F}^*$, where $\mathbb{F}^*$ denotes the multiplicative subgroup of $\mathbb{F}$, and two vertices (A, a) and (B, b) are adjacent if and only if $N(A + B) = ab$.¹

^{*}Freie Universität Berlin

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[‡]Institute for Computer Science and Control, Hungarian Academy of Sciences and BME

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¹For technical reasons we allow the two vertices to be the same, i.e. we allow loop edges.

Projective norm graphs over finite fields were introduced by Alon, Rónyai and Szabó [1] in connection with the Turán problem for complete bipartite graphs. For a prime power $q = p^k$ we use the standard notation $\mathbb{F}_q$ for the finite field with q elements. When $\mathbb{F} = \mathbb{F}_q$, then $\mathbb{K} = \mathbb{F}_{q^{t-1}}$ and the automorphism ϕ above can be chosen to be the Frobenius automorphism $\phi(X) = X^q$ and the graph $\text{NG}(\mathbb{F}_q, \mathbb{F}_{q^{t-1}})$ is denoted by $\text{NG}(q, t)$. It is not difficult to show that $\text{NG}(q, t)$ has $\frac{1}{2}q^{t-1}(q-1)(q^{t-1}-1) \approx \frac{1}{2}n^{2-\frac{1}{t}}$ edges, where $n = q^{t-1}(q-1)$ is the number of vertices. In [1] it was also proved that $\text{NG}(q, t)$ does not contain $K_{t, (t-1)!+1}$ as a subgraph, and hence has essentially the highest number of edges among graphs with this property, on the same number of vertices.

Determining the largest number $\text{ex}(n, H)$ of edges a graph on n vertices without a subgraph isomorphic to H can have is one of the classic problems of extremal graph theory, its history going back more than a century, to the theorem of Mantel about triangle-free graphs. The value of $\text{ex}(n, H)$ is settled asymptotically when H is non-bipartite, but for bipartite graphs its order of magnitude is known only in a handful of cases. Even for the simplest bipartite graphs, such as even cycles and complete bipartite graphs, the question is wide open. The general upper bound of Kővári, Sós and Turán [18] states that $\text{ex}(n, K_{t,s}) \leq c_s n^{2-\frac{1}{t}}$ for every $t \leq s$. Matching lower bounds are known only for $t = 2$ (Klein [10]), $t = 3$ (Brown [7]), and by the projective norm graphs for arbitrary $t \geq 2$ and $s \geq (t-1)! + 1$. The fundamental question of the order of magnitude of $\text{ex}(n, K_{t,s})$ is very much open for any $4 \leq t \leq s \leq (t-1)!$.

In general it is not known how large complete bipartite graphs the projective norm graph $\text{NG}(q, t)$ contains. For $t \geq 4$ it could even be the case that there is an infinite sequence of prime powers q such that $\text{NG}(q, t)$ does not contain a copy of $K_{t,t}$ and hence resolves the question of the order of magnitude of $\text{ex}(n, K_{t,t})$ for every t and s . With that, the determination of the largest integer $s(t)$, for which $\text{NG}(q, t)$ contains a $K_{t,s(t)}$ for every large enough prime power q , has potentially far reaching consequences. For $t = 2$ and $t = 3$ the projective norm graph $\text{NG}(q, t)$ does contain a $K_{t, (t-1)!}$ by combinatorial reasons (the KST upper bound), so $s(2) = 1$ and $s(3) = 2$. For $t \geq 4$ however, it was only known that $t-1 \leq s(t) \leq (t-1)!$.

In this direction Grosu [14] has recently shown that $\text{NG}(p, 4)$ contains a copy of the complete bipartite graph $K_{4,6}$ for roughly $\frac{1}{9}$ -fraction of all primes p . In [2], among other things, this was extended for any prime power q if the characteristic p is not 2 or 3. Furthermore, the proof also provided many copies of $K_{4,6}$. Computer calculations have also suggested that the same holds in the case $p \in \{2, 3\}$ as well, but the arguments in [2] crucially used the restriction on the characteristic. In this paper we provide different arguments to show the existence of $K_{4,6}$ in $\text{NG}(q, 4)$ for the cases when $p \not\equiv 1 \pmod{3}$ and $q \geq 5$, and hence establish $s(4) = 6$. In the process we uncover a close connection between the norm equation systems arising from the projective norm graph and the classic Singer difference sets. We consider this connection one of the main contributions of our paper. In the next subsection we introduce the necessary background for the latter.

1.2 Difference sets

Given a multiplicative group $\mathcal{G}$, a subset $\mathcal{D} \subseteq \mathcal{G}$ is called a *planar difference set* if every non-identity element $A \in \mathcal{G}$ has a unique representation as a product of an element from $\mathcal{D}$ and an element from $\mathcal{D}^{-1}$, where $\mathcal{D}^{-1} := \{d^{-1} : d \in \mathcal{D}\}$ denotes the set of inverses of the elements of $\mathcal{D}$. We refer to this representation as the *mixed representation of A with respect to $\mathcal{D}$* .

Difference sets in finite groups are central and diverse objects in design theory with a rich history and numerous applications both inside and outside mathematics. For a gentle introduction and survey the reader may consult e.g. [21]. If a finite group $\mathcal{G}$ admits a planar difference set of size m , then its order, by simple counting, must be of the form $\ell^2 + \ell + 1$, where $\ell = m - 1$. Planar difference sets in Abelian groups are only known to exist if $\mathcal{G}$ is a cyclic group and ℓ is a prime power. In what follows we will simply write 'difference set' instead of planar difference set. The first construction was given by Singer [25], using the finite projective plane $\text{PG}(2, q)$. A *collineation* is a one-to-one mapping c on the points of the plane carrying lines into lines. Singer proved that in $\text{PG}(2, q)$ there is always a collineation c that cyclically permutes the $q^2 + q + 1$ points. Then if we label the points P_i , $0 \leq i \leq q^2 + q$ so that $P_i = c^i(P_0)$ for every i , then the indices corresponding to points on the same line will form a difference set of size $q + 1$ in the additive cyclic group $\mathbb{Z}_{q^2+q+1}$. For the other direction he remarks that such a difference set naturally induces a projective geometry of order q . This strong connection motivates the name 'planar' difference set.

In general, for multiplicative groups $\mathcal{G}_1, \mathcal{G}_2$ two difference sets $\mathcal{D}_1 \subset \mathcal{G}_1$ and $\mathcal{D}_2 \subset \mathcal{G}_2$ are called *equivalent* if there exists a group isomorphism $\varphi : \mathcal{G}_1 \rightarrow \mathcal{G}_2$ and an element $\Gamma \in \mathcal{G}_2$ such that $\varphi(\mathcal{D}_1) = \Gamma \cdot \mathcal{D}_2$. For example, in Abelian groups any difference set $\mathcal{D}$ is equivalent to its inverse $\mathcal{D}^{-1}$ via the isomorphism $Y \rightarrow \frac{1}{Y}$. In Singer's construction from above, choosing different lines for the same collineation also results in equivalent difference sets. Singer conjectured that every difference set in $\mathbb{Z}_{q^2+q+1}$ is equivalent to his construction. This conjecture is still very much open. Berman [3] and Halberstam and Laxton [15], verifying a related conjecture of Singer, determined the exact number of reduced difference sets of Singer type in $\mathbb{Z}_{q^2+q+1}$, where a(n additive) difference set is called *reduced* if it contains both 0 and $1 \in \mathbb{Z}_{q^2+q+1}$.

We finish this subsection with an equivalent formulation of Singer's construction that will be useful later. We consider the cyclic group $\mathcal{G} = \mathbb{F}_{q^3}^* / \mathbb{F}_q^*$. The order of this group is $q^2 + q + 1$ and the cosets of those elements $A \in \mathbb{F}_{q^3}^*$ for which the trace $\text{Tr}(A) = A + A^q + A^{q^2} = 0$ form a difference set of size $q + 1$ which is equivalent to the Singer difference set (see e.g. [23]).

1.3 Results

Difference sets. In order to treat infinite difference sets as well, we introduce our definitions and results for arbitrary fields, restricting to finite fields only when necessary. This general approach also keeps the arguments more

transparent.

Let $\mathbb{F}$ be an arbitrary field and $\mathbb{K}$ a Galois extension of degree 3. Let ϕ be a nonidentity $\mathbb{F}$ -automorphism of $\mathbb{K}$. Then the Galois group of the extension $\mathbb{K}$ over $\mathbb{F}$ is $\{\phi, \psi, id\}$, where $\psi = \phi \circ \phi$ and, of course, $id = \phi \circ \phi \circ \phi$. We denote by $N_{\mathbb{K}/\mathbb{F}}$ the norm of this extension: for $A \in \mathbb{K}$ we have $N_{\mathbb{K}/\mathbb{F}}(A) = A \cdot \phi(A) \cdot \psi(A) \in \mathbb{F}$. When it causes no confusion, which will be the case most of the time, we omit writing the index $\mathbb{K}/\mathbb{F}$. Examples of such extensions are simplest cubic fields [24], which are important and well studied objects in algebraic number theory.

We shall consider two functions $h_1, h_2 : \mathbb{K} \rightarrow \mathbb{K}$ defined as

$$h_1(X) = \phi(X) \cdot X + X + 1 \quad \text{and} \quad h_2(X) = \phi(X) \cdot X + \phi(X) + 1.$$

For $i = 1, 2$ let $\mathcal{H}_i$ denote the set of roots of h_i in $\mathbb{K}$ and let $\mathcal{N}$ denote the set of elements in $\mathbb{K}^*$ with norm 1. It is easy to see that $\mathcal{N}$ is a subgroup of the multiplicative group $\mathbb{K}^*$. In our first result we prove that $\mathcal{H}_1$ and $\mathcal{H}_2$ form a difference set in the cyclic group $\mathcal{N}$ with an explicit formula for the mixed representation.

Theorem 1.1. *The sets $\mathcal{H}_1$ and $\mathcal{H}_2$ are equivalent difference sets in the group $\mathcal{N}$ and $\mathcal{H}_2 = \mathcal{H}_1^{-1}$. Furthermore, the unique mixed representation of an element $A \in \mathcal{N} \setminus \{1\}$ (with respect to $\mathcal{H}_i$) is given by the following explicit formulas:*

$$A_1 = \frac{A \cdot \phi(A) - 1}{1 - \phi(A)} \in \mathcal{H}_1 \quad \text{and} \quad A_2 = \frac{A - A \cdot \phi(A)}{A \cdot \phi(A) - 1} \in \mathcal{H}_2. \quad (1)$$

Infinite difference sets were earlier constructed by Hughes [16] using a greedy-like approach. Our construction is more explicit and so offers more possibilities to study these nice combinatorial structures.

Next we spell out the statement of our theorem for finite fields. We shall show that in this case our difference sets are of Singer type. Let $\mathbb{F} = \mathbb{F}_q$, $\mathbb{K} = \mathbb{F}_{q^3}$ and take ϕ to be the Frobenius automorphism $X \rightarrow X^q$. Then $h_1(X)$ and $h_2(X)$ become polynomials over $\mathbb{F}_q$, namely

$$h_1(X) = X^{q+1} + X + 1 \quad \text{and} \quad h_2(X) = X^{q+1} + X^q + 1,$$

and the group $\mathcal{N}$ is the unique subgroup of order $q^2 + q + 1$ in $\mathbb{F}_{q^3}^*$.

Corollary 1.2. *The root set $\mathcal{H}_1 \subseteq \mathbb{F}_{q^3}$ of the polynomial $h_1(X) = X^{q+1} + X + 1 \in \mathbb{F}_q[X]$ and the root set $\mathcal{H}_2 \subseteq \mathbb{F}_{q^3}$ of the polynomial $h_2(X) = X^{q+1} + X^q + 1 \in \mathbb{F}_q[X]$ are equivalent difference sets of size $q + 1$ in the order $q^2 + q + 1$ cyclic group $\mathcal{N}$ of norm 1 elements of $\mathbb{F}_{q^3}$ with $\mathcal{H}_2 = \mathcal{H}_1^{-1}$. Furthermore the unique mixed representation of an element $A \in \mathcal{N} \setminus \{1\}$ (with respect to $\mathcal{H}_i$) is given by the following explicit formulas:*

$$A_1 = \frac{A^{q+1} - 1}{1 - A^q} \in \mathcal{H}_1 \quad \text{and} \quad A_2 = \frac{A - A^{q+1}}{A^{q+1} - 1} \in \mathcal{H}_2.$$

Moreover, both $\mathcal{H}_1$ and $\mathcal{H}_2$ are equivalent to the Singer difference set.

Even though the difference sets given here are equivalent to Singer's construction, their description as the roots of a simple polynomial and the explicit formulas for the mixed representation are interesting on their own right.

In Section 3 we generalize Theorem 1.1 and Corollary 1.2 from the planar case to Singer difference sets of arbitrary classical parameters.

Projective norm graphs. As already mentioned earlier, one of the main results in [2] is that if $p > 3$ then $\text{NG}(q, 4)$ contains $K_{4,6}$ as a subgraph. Relying heavily on the properties of the Singer difference set from Corollary 1.2 and its connection to norm equation systems, here we settle the remaining cases of characteristic 2 and 3.

Theorem 1.3. *Let $q = p^k > 4$ where $p \not\equiv 1 \pmod{3}$ is a prime. Then $\text{NG}(q, 4)$ contains $K_{4,6}$ as a subgraph. In particular, $\text{NG}(2^k, 4)$ and $\text{NG}(3^\ell, 4)$ contain $K_{4,6}$ as a subgraph for $k \geq 3$ and $\ell \geq 2$.*

Complementing Theorem 1.3 we remark that it is immediate that $\text{NG}(2, 4)$ does not contain $K_{4,6}$, and in $\text{NG}(3, 4)$ and $\text{NG}(4, 4)$ we verified by computer search that there is no $K_{4,6}$ either.

2 Proofs

2.1 Properties of the sets $\mathcal{H}_i$

We start by proving a few helpful properties of the sets $\mathcal{H}_i$, including their unique mixed representation property from Theorem 1.1.

Proposition 2.1.

- (i) For $Y \in \mathbb{K}^*$ we have $h_2\left(\frac{1}{Y}\right) = \frac{h_1(Y)}{\Phi(Y) \cdot Y}$. In particular, $\mathcal{H}_2 = \mathcal{H}_1^{-1}$.
- (ii) Every $A \in \mathcal{N} \setminus \{1\}$ can be represented uniquely as a product $A = A_1 \cdot A_2$ of an element A_1 of $\mathcal{H}_1$ and an element A_2 of $\mathcal{H}_2$. This representation is given by

$$A_1 = \frac{A \cdot \phi(A) - 1}{1 - \phi(A)} \quad \text{and} \quad A_2 = \frac{A - A \cdot \phi(A)}{A \cdot \phi(A) - 1}. \quad (2)$$

(iii)

$$\mathcal{H}_1 \cap \mathcal{H}_2 = \mathcal{H}_1 \cap \mathbb{F} = \mathcal{H}_2 \cap \mathbb{F} = \{Y \in \mathbb{F} : Y^2 + Y + 1 = 0\},$$

in particular, if $\mathbb{F} = \mathbb{F}_q$, $\mathbb{K} = \mathbb{F}_{q^3}$ then

$$\mathcal{H}_1 \cap \mathcal{H}_2 = \begin{cases} \{1\} & \text{if } q \equiv 0 \pmod{3} \\ \{\alpha, \alpha^{-1}\}, \text{ where } \alpha^3 = 1, \alpha \neq 1 & \text{if } q \equiv 1 \pmod{3} \\ \emptyset & \text{if } q \equiv 2 \pmod{3} \end{cases}.$$

- (iv) If $Y \in \mathbb{K}$ then $\phi(h_i(Y)) = h_i(\phi(Y))$, in particular $\phi(\mathcal{H}_i) = \mathcal{H}_i$.

Proof. (i) For $Y \in \mathbb{K}$ we have

$$\begin{aligned} h_2\left(\frac{1}{Y}\right) &= \frac{1}{Y} \cdot \phi\left(\frac{1}{Y}\right) + \phi\left(\frac{1}{Y}\right) + 1 = \frac{1}{Y \cdot \phi(Y)} + \frac{1}{\phi(Y)} + 1 = \\ &= \frac{1 + Y + Y \cdot \phi(Y)}{Y \cdot \phi(Y)} = \frac{h_1(Y)}{Y \cdot \phi(Y)}. \end{aligned}$$

The rest of the statement follows by the definition of the $\mathcal{H}_i$'s.

(ii) First we show the uniqueness of the representation of the form (2). Suppose that $A \in \mathcal{N} \setminus \{1\}$ and $A = A_1 \cdot A_2$ with $A_1 \in \mathcal{H}_1$ and $A_2 = A/A_1 \in \mathcal{H}_2$. Then

$$\begin{aligned} A_1 \cdot \phi(A_1) + A_1 + 1 &= 0, \\ \frac{A}{A_1} \cdot \phi\left(\frac{A}{A_1}\right) + \phi\left(\frac{A}{A_1}\right) + 1 &= \frac{A}{A_1} \cdot \frac{\phi(A)}{\phi(A_1)} + \frac{\phi(A)}{\phi(A_1)} + 1 = 0. \end{aligned}$$

By expressing $\phi(A_1)$ from the two equations we obtain

$$\frac{A_1 + 1}{A_1} = \frac{A \cdot \phi(A)}{A_1} + \phi(A).$$

Solving for A_1 gives that the only possibility is

$$A_1 = \frac{A \cdot \phi(A) - 1}{1 - \phi(A)}.$$

This gives uniqueness and the stated formula (1) for $A_2 = A/A_1$ as well. It remains to verify that $A_i \in \mathcal{H}_i$. We consider first A_1 . Using

$$N(A) = A \cdot \phi(A) \cdot \psi(A) = 1$$

we have

$$\begin{aligned} h_1(A_1) &= \frac{A \cdot \phi(A) - 1}{1 - \phi(A)} \cdot \phi\left(\frac{A \cdot \phi(A) - 1}{1 - \phi(A)}\right) + \frac{A \cdot \phi(A) - 1}{1 - \phi(A)} + 1 = \\ &= \frac{A \cdot \phi(A) - 1}{1 - \phi(A)} \cdot \frac{\phi(A) \cdot \psi(A) - 1}{1 - \psi(A)} + \frac{\phi(A)(A - 1)}{1 - \phi(A)} \\ &= \frac{\frac{1}{\psi(A)} - 1}{1 - \phi(A)} \cdot \frac{\frac{1}{A} - 1}{1 - \psi(A)} + \frac{\frac{1}{A \cdot \psi(A)}(A - 1)}{1 - \phi(A)} = \\ &= \frac{1 - A}{A \cdot \psi(A) \cdot (1 - \phi(A))} + \frac{A - 1}{A \cdot \psi(A) \cdot (1 - \phi(A))} = 0, \end{aligned}$$

and hence $A \in \mathcal{H}_1$. The verification of $A_2 \in \mathcal{H}_2$ is a similar calculation that we leave to the reader.

(iii) To see the desired equalities of sets note that for an element $Y \in \mathbb{K}$ we have $h_1(Y) = h_2(Y)$ if and only if $Y = \phi(Y)$, which happens if and only if $Y \in \mathbb{F}$.

If furthermore $\mathbb{F} = \mathbb{F}_q$, then for $3|q$ we have $X^2 + X + 1 = (X - 1)^2$, while otherwise the non-trivial third-roots of unity are in $\mathbb{F}$ if and only if $3|q - 1$.

(iv) The statement is a direct consequence of the fact that ϕ is an automorphism that fixes $\mathbb{F}$. $\square$

2.2 Difference sets and a norm equation system

After Proposition 2.1(ii), all we need in order to complete the proof of Theorem 1.1 is to show that $\mathcal{H}_1, \mathcal{H}_2 \subseteq \mathcal{N}$. In our next result we are not only proving that the elements of $\mathcal{H}_1 \cup \mathcal{H}_2$ indeed have norm 1, but we are also able to characterize them as the solutions in $\mathbb{K}$ of the norm equation system

$$N(X) = 1, \quad N(X + 1) = -1, \quad (3)$$

which is closely related to projective norm graphs. Besides this connection, the system (3) also arises naturally in algebraic number theory. When $\mathbb{K}$ is a number field, then the algebraic integer solutions of (3) will be exceptional units in the sense of Nagell [22]. Exceptional units are interesting objects, in particular one can show that their number is always finite.

Proposition 2.2. *$Y \in \mathbb{K}$ is a solution of (3) if and only if $Y \in \mathcal{H}_1 \cup \mathcal{H}_2$.*

Proof. The proof is adapted from [2] which handled the finite field setting. We first write the equations of (3) in simple product form.

$$\begin{aligned} X \cdot \phi(X) \cdot \psi(X) &= 1 \\ (X + 1) \cdot \phi(X + 1) \cdot \psi(X + 1) &= (X + 1) \cdot (\phi(X) + 1) \cdot (\psi(X) + 1) = -1 \end{aligned}$$

Expressing $\psi(X)$ from both equations and clearing denominators gives

$$(X + 1) \cdot (\phi(X) + 1) = -X \cdot \phi(X) - X \cdot \phi(X) \cdot (X + 1) \cdot (\phi(X) + 1).$$

After expanding and simplifying we obtain

$$X^2 \cdot (\phi(X))^2 + X \cdot (\phi(X))^2 + X^2 \cdot \phi(X) + 3X \cdot \phi(X) + \phi(X) + X + 1 = 0. \quad (4)$$

It is easy to check that the left hand side of (4) is

$$h_1(X) \cdot h_2(X) = (\phi(X) \cdot X + X + 1) \cdot (X \cdot \phi(X) + \phi(X) + 1).$$

Clearly, by the above calculations, any solution Y of (3) will be a solution of (4), hence $h_1(Y) \cdot h_2(Y) = 0$, which in turn is equivalent to $Y \in \mathcal{H}_1 \cup \mathcal{H}_2$.

Conversely, let $Y \in \mathcal{H}_1 \cup \mathcal{H}_2$ i.e. $h_1(Y) \cdot h_2(Y) = 0$. Then either

$$h_1(Y) = 0, \text{ in which case } \phi(Y) = -\frac{Y+1}{Y} =: u(Y),$$

or

$$h_2(Y) = 0, \text{ in which case } \phi(Y) = -\frac{1}{Y+1} =: v(Y).$$

Note that the denominator is non-zero in both cases. Using part (iv) of Proposition 2.1, in the first case we obtain

$$\psi(Y) = \phi \circ \phi(Y) = u(u(Y)) = -\frac{-\frac{Y+1}{Y} + 1}{-\frac{Y+1}{Y}} = v(Y),$$

while in the second case we get

$$\psi(Y) = \phi \circ \phi(Y) = v(v(Y)) = -\frac{1}{-\frac{1}{Y+1} + 1} = u(Y).$$

In both cases we have $\{\phi(Y), \psi(Y)\} = \{u(Y), v(Y)\}$ and therefore

$$N(Y) = Y \cdot \phi(Y) \cdot \psi(Y) = Y \cdot u(Y) \cdot v(Y) = Y \cdot \left(-\frac{1}{Y+1}\right) \cdot \left(-\frac{Y+1}{Y}\right) = 1.$$

Similarly, for the norm of $Y+1$ we obtain

$$\begin{aligned} N(Y+1) &= (Y+1) \cdot \phi(Y+1) \cdot \psi(Y+1) = (Y+1) \cdot (\phi(Y)+1) \cdot (\psi(Y)+1) \\ &= (Y+1) \cdot (u(Y)+1) \cdot (v(Y)+1) \\ &= (Y+1) \cdot \left(-\frac{Y+1}{Y} + 1\right) \cdot \left(-\frac{1}{Y+1} + 1\right) = -1. \end{aligned}$$

This finishes the proof. $\square$

Proof of Theorem 1.1. Proposition 2.2 shows that $N(X) = 1$ for every $X \in \mathcal{H}_1 \cup \mathcal{H}_2$, hence $\mathcal{H}_1, \mathcal{H}_2 \subseteq \mathcal{N}$. The equality $\mathcal{H}_2 = \mathcal{H}_1^{-1}$ is just part (i) of Proposition 2.1, while the unique mixed representation of an element $A \in \mathcal{N} \setminus \{1\}$ is provided by part (ii) of Proposition 2.1. $\square$

Proof of Corollary 1.2. Most of the statements follow directly by applying Theorem 1.1 to the setting $\mathbb{F} = \mathbb{F}_q$, $\mathbb{K} = \mathbb{F}_{q^3}$, and $\phi(X) = X^q$. Note that then $\mathcal{N}$ is an order $q^2 + q + 1$ multiplicative group, hence any difference set in it, in particular $\mathcal{H}_1$ and $\mathcal{H}_2$, must have size $q+1$. In particular, this implies that both h_1 and h_2 have all their $q+1$ roots in $\mathcal{N} \subseteq \mathbb{F}_{q^3}$.

It remains to prove the equivalence of $\mathcal{H}_1$ (and so of $\mathcal{H}_2$) to the Singer difference set $\mathcal{S}$, for which we use the description mentioned at the end of Subsection 1.2, i.e.

$$\mathcal{S} = \{A \cdot \mathbb{F}_q^* : A \in \mathbb{F}_{q^3}^*, \text{Tr}(A) = 0\} \subseteq \mathbb{F}_{q^3}^* / \mathbb{F}_q^*.$$

Consider the group homomorphism $\Phi : \mathbb{F}_{q^3}^* \rightarrow \mathbb{F}_{q^3}^*$, given by $\Phi(A) = A^{q-1}$. Since $N(\phi(A)) = A^{q^3-1} = 1$, the image of Φ is contained in $\mathcal{N}$. The kernel is $\mathbb{F}^*$ and $|\mathcal{N}| = q^2 + q + 1 = \frac{q^3-1}{q-1}$, so the quotient map $\bar{\Phi}$ provides an isomorphism between the groups $\mathbb{F}_{q^3}^* / \mathbb{F}_q^*$ and $\mathcal{N}$.

We claim that $\bar{\Phi}$ maps $\mathcal{S}$ into $\mathcal{H}_1$, i.e. if $A \in \mathbb{F}_{q^3}^*$ is such that $\text{Tr}(A) = A^{q^2} + A^q + A = 0$, then $A^{q-1} \in \mathcal{H}_1$. Indeed,

$$\begin{aligned} h_1(A^{q-1}) &= A^{(q-1)(q+1)} + A^{q-1} + 1 = \frac{A^{q^2}}{A} + A^{q-1} + 1 = \\ &= \frac{(-A^q - A)}{A} + A^{q-1} + 1 = -A^{q-1} - 1 + A^{q-1} + 1 = 0. \end{aligned}$$

Using the injectivity of $\bar{\Phi}$ and $|\mathcal{S}| = |\mathcal{H}_1|$, we obtain that $\bar{\Phi}(\mathcal{S}) = \mathcal{H}_1$, showing that $\bar{\Phi}$ gives the desired equivalence of the difference sets $\mathcal{S}$ and $\mathcal{H}_1$. $\square$

2.3 Norm equation systems with the maximum number of solutions

Let $\mathbb{F}$ be an arbitrary field and $\mathbb{K}$ a Galois extension of degree 3. Finding a $K_{4,6}$ in $\text{NG}(\mathbb{F}, \mathbb{K})$, up to a couple of technicalities to be handled later, is essentially equivalent to finding distinct pairs $(A_1, a_1), (A_2, a_2), (A_3, a_3) \in \mathbb{K} \times \mathbb{F}^*$ such that the system

$$N(X + A_1) = a_1, \quad \text{NG}(X + A_2) = a_2, \quad N(X + A_3) = a_3, \quad (5)$$

has six solutions $X \in \mathbb{K}$. Note that by the $K_{4,7}$ -freeness of the projective norm graph $\text{NG}(\mathbb{F}, \mathbb{K})$ (see Subsection 5.1 in the Appendix), we also know that (5) can have *at most* six solutions for any values of the parameters.

In what follows we will study this system for special sets of parameters and try to find particular choices where the maximum possible six solutions are attained. The main result of this subsection is that in some cases this is indeed possible. For a fixed element $A \in \mathcal{N} \setminus \{1\}$ consider the following system of norm equations:

$$N(X) = 1, \quad N(X + 1) = -1, \quad N(X + A) = -1. \quad (6)$$

Theorem 2.3. *Let $q = p^k > 4$, where $p \not\equiv 1 \pmod{3}$ is prime, and let $\mathbb{F} = \mathbb{F}_q$, $\mathbb{K} = \mathbb{F}_{q^3}$. Then there exists an element $A \in \mathcal{N} \setminus \{1\}$ such that the system (6) has 6 solutions in $\mathbb{F}_{q^3}$.*

The starting point in the proof of Theorem 2.3 is an observation, valid over arbitrary fields, that connects the solution set of the system (6) to the solution set of (3) and hence to the difference sets $\mathcal{H}_1, \mathcal{H}_2$.

Proposition 2.4. *An element $Y \in \mathbb{K}$ is a solution of (6) with parameter $A \in \mathcal{N} \setminus \{1\}$ if and only if Y and $\frac{A}{Y}$ are both contained in $\mathcal{H}_1 \cup \mathcal{H}_2$.*

Proof. By Proposition 2.2 Y and $\frac{A}{Y}$ are both contained in $\mathcal{H}_1 \cup \mathcal{H}_2$ if and only if they are both solutions of (3). We show that this is equivalent to Y being a solution of (6).

Suppose first that $Y \in \mathbb{K}$ is a solution of (6). Then a fortiori Y is a solution of (3) and $Y \neq 0$. Also,

$$\begin{aligned} N\left(\frac{A}{Y}\right) &= \frac{N(A)}{N(Y)} = \frac{1}{1} = 1, \\ N\left(\frac{A}{Y} + 1\right) &= N\left(\frac{A + Y}{Y}\right) = \frac{N(A + Y)}{N(Y)} = \frac{-1}{1} = -1, \end{aligned}$$

hence $\frac{A}{Y}$ is also a solution of (3).

Conversely, assume that Y and $\frac{A}{Y}$ are both solutions of (3). Then, in particular, Y satisfies the first two equations from (6), as for the third one we have

$$N(Y + A) = N\left(Y\left(1 + \frac{A}{Y}\right)\right) = N(Y)N\left(\frac{A}{Y} + 1\right) = 1 \cdot -1 = -1,$$

and hence Y is a solution of (6). $\square$

By the previous proposition we will be looking for product representations $Y \cdot \frac{A}{Y}$ of an element A from the set $\mathcal{H}_1 \cup \mathcal{H}_2$. To prove Theorem 2.3 we actually need to find an element $A \in \mathcal{N} \setminus \{1\}$ that has three such product representations $A = B_1 C_1 = B_2 C_2 = B_3 C_3$, such that the six elements $B_1, C_1, B_2, C_2, B_3, C_3 \in \mathcal{H}_1 \cup \mathcal{H}_2$ are all distinct. For this we will crucially use that $\mathcal{H}_1$ and $\mathcal{H}_2 = \mathcal{H}_1^{-1}$ are difference sets in $\mathcal{N}$ and inverses of each other.

Recall that if $\mathcal{D}$ is any difference set in some multiplicative group $\mathcal{G}$ then every element $A \in \mathcal{G} \setminus \{1\}$ has a unique representation, called mixed representation, as a product $B \cdot C = A$ such that one of B and C is from $\mathcal{D}$ and the other is from $\mathcal{D}^{-1}$. In the next propositions we summarize our knowledge about other product representations. To this end we will call a product $B \cdot C = A$ a $\mathcal{D}$ -representation of the element $A \in \mathcal{G}$ if both B and C are from $\mathcal{D}$.

Proposition 2.5. *Let $\mathcal{D}$ be an arbitrary difference set in some multiplicative group $\mathcal{G}$. Then every $A \in \mathcal{G}$ has at most one $\mathcal{D}$ -representation.*

Proof. Let us assume that $D_1 D_2 = D_3 D_4$ for some $D_1, D_2, D_3, D_4 \in \mathcal{D}$. Then $D_1 D_3^{-1} = D_4 D_2^{-1}$, and this, by the difference set property, is either 1 or we have $D_1 = D_4$ and $D_2 = D_3$: in any case $\{D_1, D_2\} = \{D_3, D_4\}$. $\square$

The explicit descriptions of our difference sets allow us to characterize when an $\mathcal{H}_i$ -representation with distinct factors exists.

Proposition 2.6.

- (i) *If $\text{char}(\mathbb{F}) \neq 2$ then $A \in \mathcal{N}$ has an $\mathcal{H}_1$ -representation with different factors if and only if $(A + 1 - A\phi(A))^2 - 4A$ is a non-zero square in $\mathbb{K}$.*
- (ii) *If $\text{char}(\mathbb{F}) \neq 2$ then $A \in \mathcal{N}$ has an $\mathcal{H}_2$ -representation with different factors if and only if $(A\phi(A) + \phi(A) - 1)^2 - 4A\phi(A)$ is a non-zero square in $\mathbb{K}$.*

Proof. In the case of finite fields most parts of the argument have already appeared in [2]. As the proof of the two parts is analogous, below we only present the one of (i).

Suppose first that $A \in \mathcal{N}$ has an $\mathcal{H}_1$ -representation. This means that there is an element $Y \in \mathbb{K}$ such that Y and $\frac{A}{Y}$ are both roots of h_1 , i.e.

$$h_1(Y) = \phi(Y) \cdot Y + Y + 1 = 0 \quad \text{and} \quad h_1\left(\frac{A}{Y}\right) = \phi\left(\frac{A}{Y}\right) \cdot \frac{A}{Y} + \frac{A}{Y} + 1 = 0.$$

after expressing $\phi(Y)$ from both equations, letting them being equal and clearing denominators we obtain $Y^2 + (A + 1 - A\phi(A)) \cdot Y + A = 0$. Clearly, the role of Y and $\frac{A}{Y}$ can be switched, which means that both Y and $\frac{A}{Y}$ are roots of the quadratic equation

$$X^2 + (A + 1 - A\phi(A)) \cdot X + A = 0. \tag{7}$$

If $\text{char}(\mathbb{F}) \neq 2$, this is possible only if the discriminant

$$D = (A + 1 - A\phi(A))^2 - 4A$$

is a nonzero square in $\mathbb{K}$.

For the other direction suppose that $D = (A + 1 - A\phi(A))^2 - 4A$ is a nonzero square in $\mathbb{K}$, i.e. there is some element $G \in \mathbb{K}^*$ such that $D = G^2$. Then we know that the quadratic equation in (7) has two different roots, namely $X_{\pm} = \frac{A\phi(A) - A - 1 \pm G}{2}$. Clearly, $X_+ \cdot X_- = A$, so to finish the proof it is enough to show that $X_{\pm} \in \mathcal{H}_1$, i.e. $h_1(X_{\pm}) = 0$.

Using $N(A) = A\phi(A)\psi(A) = 1$ we have

$$\begin{aligned} \phi(D) &= \phi\left((A + 1 - A\phi(A))^2 - 4A\right) = (\phi(A) + 1 - \phi(A)\psi(A))^2 - 4\phi(A) = \\ &= \left(\phi(A) + 1 - \frac{1}{A}\right)^2 - 4\phi(A) = \frac{1}{A^2} \left((A\phi(A) + A - 1)^2 - 4A^2\phi(A)\right) = \\ &= \frac{1}{A^2} \left((A + 1 - A\phi(A))^2 - 4A\right) = \frac{1}{A^2} D. \end{aligned}$$

Then $\left(\frac{\phi(G)}{G}\right)^2 = \frac{\phi(G^2)}{G^2} = \frac{\phi(D)}{D} = \frac{\frac{1}{A^2}D}{D} = \frac{1}{A^2}$ and hence $\frac{\phi(G)}{G} = \pm \frac{1}{A}$. However $N\left(\frac{\phi(G)}{G}\right) = \frac{\phi(G)}{G} \cdot \phi\left(\frac{\phi(G)}{G}\right) \cdot \psi\left(\frac{\phi(G)}{G}\right) = \frac{\phi(G)}{G} \cdot \frac{\psi(G)}{\phi(G)} \cdot \frac{G}{\psi(G)} = 1$ which, as $\text{char}(\mathbb{F}) \neq 2$, excludes $\frac{\phi(G)}{G} = -\frac{1}{A}$. Therefore $\phi(G) = \frac{1}{A}G$. As a consequence we get

$$\begin{aligned} \phi(X_{\pm}) &= \phi\left(\frac{A\phi(A) - A - 1 \pm G}{2}\right) = \frac{\phi(A)\psi(A) - \phi(A) - 1 \pm \phi(G)}{2} = \\ &= \frac{\frac{1}{A} - \phi(A) - 1 \pm \frac{1}{A}G}{2} = \frac{1}{A}X_{\pm} + \frac{1 - A\phi(A)}{A}. \end{aligned}$$

Now we are ready to substitute $X_{\pm}$ into h_1 .

$$\begin{aligned} h_1(X_{\pm}) &= X_{\pm} \cdot \phi(X_{\pm}) + X_{\pm} + 1 = X_{\pm} \cdot \left(\frac{1}{A}X_{\pm} + \frac{1 - A\phi(A)}{A}\right) + X_{\pm} + 1 = \\ &= \frac{1}{A} (X_{\pm}^2 + (A + 1 - A\phi(A)) \cdot X_{\pm} + A) = 0, \end{aligned}$$

where at the last equality we just used that $X_{\pm}$ are the roots of (7). $\square$

For the elements of $\mathcal{H}_i$ the existence of a $\mathcal{H}_{3-i}$ -representation follows directly.

Proposition 2.7. *If $A \in \mathcal{H}_i$ then its unique $\mathcal{H}_{3-i}$ -representation is given by $\frac{1}{\phi(A)} \cdot \frac{1}{\psi(A)} = A$.*

Proof. On the one hand, as $A \in \mathcal{H}_i \subseteq \mathcal{N}$, we have that $A = \frac{N(A)}{\phi(A) \cdot \psi(A)} = \frac{1}{\phi(A)} \cdot \frac{1}{\psi(A)}$. On the other hand, by Proposition 2.1(i) we have $\frac{1}{A} \in \mathcal{H}_{3-i}$, and so by Proposition 2.1(iv) we have $\frac{1}{\phi(A)} = \phi\left(\frac{1}{A}\right) \in \mathcal{H}_{3-i}$ and $\frac{1}{\psi(A)} = \phi\left(\frac{1}{\phi(A)}\right) \in \mathcal{H}_{3-i}$. The uniqueness follows from Proposition 2.5. $\square$

From now on we will consider the special case $\mathbb{F} = \mathbb{F}_q$ (and $\mathbb{K} = \mathbb{F}_{q^3}$), and separate the cases according to the characteristic modulo 3.

2.3.1 Characteristic 2

In this subsection we settle the case of characteristic 2 which was left open in [2]. Our method extends to odd characteristic $p \equiv 2 \pmod{3}$, which was settled, in a much stronger form, already in [2].

First we show the existence of six solutions when q is congruent to 2 modulo 3 (as opposed to p).

Proposition 2.8. *Let $q > 2$ be a prime power such that $q \equiv 2 \pmod{3}$, and let $\mathbb{F} = \mathbb{F}_q$ and $\mathbb{K} = \mathbb{F}_{q^3}$. Then there exists an element $A \in \mathcal{N} \setminus \{1\}$ such that the system (6) has 6 solutions in $\mathbb{F}_{q^3}$.*

Proof. We will find an element $A \in \mathcal{N} \setminus \{1\}$ that has an $\mathcal{H}_i$ -representation $A = B_i \cdot C_i$ for both $i = 1$ and 2 , such that $B_i \neq C_i$. These four elements, together with the two elements $A_1 \in \mathcal{H}_1$ and $A_2 \in \mathcal{H}_2$ from the unique mixed representation of A (which exists by Proposition 2.1(ii)) give us six distinct solutions of (6). Indeed, they are solutions of (6) by Proposition 2.4 and their distinctness follows immediately from the fact that the sets $\mathcal{H}_1$ and $\mathcal{H}_2$ are disjoint by Proposition 2.1(iii).

In order to find the appropriate element A , for a set $\mathcal{H} \subseteq \mathcal{N}$ we define

$$\mathcal{H}^* := \{B \cdot C : B, C \in \mathcal{H}, B \neq C\}$$

to be the set of its pairwise products from distinct factors, and show that $\mathcal{H}_1^* \cap \mathcal{H}_2^*$ is not empty.

First note that by Proposition 2.6(ii) the pairwise products of the elements of $\mathcal{H}_i$ are all distinct, hence the cardinality of $\mathcal{H}_i^*$ is $\binom{|\mathcal{H}_i|}{2} = \frac{q^2+q}{2}$. Both $\mathcal{H}_1^*$ and $\mathcal{H}_2^*$ are subsets of the $(q^2 + q)$ -element set $\mathcal{N} \setminus \{1\}$. For this note that $\mathcal{H}_i$ is contained in $\mathcal{N}$, which is closed under multiplication, and that $1 \notin \mathcal{H}_i^*$ since $\mathcal{H}_i$ is disjoint from its inverse $\mathcal{H}_{3-i}$ by our assumption on q and Proposition 2.1(iii). Hence the only way $\mathcal{H}_1^*$ and $\mathcal{H}_2^*$ could be disjoint is if their union is $\mathcal{N} \setminus \{1\}$. In this case however it would also hold that

$$\sigma(\mathcal{N} \setminus \{1\}) = \sigma(\mathcal{H}_1^*) + \sigma(\mathcal{H}_2^*), \quad (8)$$

where $\sigma(\mathcal{S})$ denotes the sum of the elements of a subset $\mathcal{S} \subseteq \mathbb{F}_{q^3}$. On the one hand the set $\mathcal{N}$ is the collection of all $q^2 + q + 1$ roots in $\mathbb{F}_{q^3}$ of the polynomial $X^{q^2+q+1} - 1$ and so $\sigma(\mathcal{N})$ is (-1) times the coefficient of X^{q^2+q} in this polynomial, which is 0. From this we obtain that $\sigma(\mathcal{N} \setminus \{1\}) = -1$. On the other hand $\mathcal{H}_i$ is the set of all $q + 1$ roots in $\mathbb{F}_{q^3}$ of the polynomial $h_i(X)$, hence $\sigma(\mathcal{H}_i^*)$ is the coefficient of X^{q-1} in $h_i(X)$, which is 0 for $q > 2$. We arrived to a contradiction, as the left hand side of (8) is (-1) , while the right hand side is 0. $\square$

Proof of Theorem 2.3 for $p \equiv 2 \pmod{3}$. First note that $\mathbb{F}_{q^6}$ is a cubic extension of $\mathbb{F}_{q^2}$ and the norm of an element $B \in \mathbb{F}_{q^3} \subseteq \mathbb{F}_{q^6}$ is the same, irrespective in which of the two fields we compute it: $N_{\mathbb{F}_{q^3}/\mathbb{F}_q}(B) = N_{\mathbb{F}_{q^6}/\mathbb{F}_{q^2}}(B)$. This means that if for an element $A \in \mathbb{F}_{q^3}$ with $N_{\mathbb{F}_{q^3}/\mathbb{F}_q}(A) = 1$ the system (6) with the norm function $N_{\mathbb{F}_{q^3}/\mathbb{F}_q}$ has six distinct solutions $X_1, \dots, X_6 \in \mathbb{F}_{q^3}$, then the very same six elements are also solutions of the the system (6) with the norm function $N_{\mathbb{F}_{q^6}/\mathbb{F}_{q^2}}$.

Now let $p \equiv 2 \pmod{3}$ be a prime and let $q = p^k = p^{\ell 2^m}$ be an arbitrary power where ℓ is odd. Then Proposition 2.8 gives the required six distinct solutions when $m = 0$ and $p^\ell > 2$. By repeated application of the above observation, the statement also follows for any positive integer m and $p^\ell > 2$. These include all the powers when $p > 2$.

When $p = 2$ then only those powers are included where $\ell \geq 3$. So we are left with prime powers of the form 2^{2^m} . To settle these last cases one first resolves the problem when the prime power is $2^{2^2} = 16$ and then uses the above squaring trick to deduce the case of arbitrary $2^{2^m} = 16^{2^{m-2}}$.

For $q = 16$ we have found the appropriate $A \in \mathbb{F}_{16^3}$ of norm 1, for which the system (6) has six distinct solution with the aid of a computer. To describe this example, let U be the primitive element of $\mathbb{F}_{16^3}^*$ whose minimal polynomial over $\mathbb{F}_2$ is $X^{12} + X^7 + X^6 + X^5 + X^3 + X + 1$, and consider the system

$$N(X) = 1, \quad N(X + 1) = 1, \quad N(X + U^{405}) = 1.$$

By Magma Calculator [6] it is easily verified that $A = U^{405}$ is in $\mathcal{N} \setminus \{1\}$, and that the system has indeed six solutions, namely $U^{1725}, U^{2775}, U^{3435} \in \mathcal{H}_1$ and $U^{1065}, U^{2130}, U^{2370} \in \mathcal{H}_2$ with

$$A = U^{1065} \cdot U^{3435} = U^{1725} \cdot U^{2775} = U^{2130} \cdot U^{2370}.$$

□

2.3.2 Characteristic 3

Proof of Theorem 2.3 for $p = 3$. Just like in the previous subsection, we will find an element $A \in \mathcal{N} \setminus \{1\}$ which has an $\mathcal{H}_i$ -representation $A = B_i \cdot C_i$ for both $i = 1$ and 2 , such that these four elements and the two elements $A_1 \in \mathcal{H}_1$ and $A_2 \in \mathcal{H}_2$ from the mixed representation of A (which exists by Proposition 2.1(ii)) are pairwise distinct and hence provide six distinct solutions of (6).

We do this in two steps. First we find an element that has both $\mathcal{H}_1$ - and $\mathcal{H}_2$ -representation, but in one of them the factors are not distinct.

Lemma 2.9. *For $i = 1$ or 2 there is an element $C \in \mathcal{H}_i \setminus \{1\}$, such that C^2 has an $\mathcal{H}_{3-i}$ -representation $C^2 = B \cdot E$ with distinct factors $B \neq E$.*

Let us fix elements $C \in \mathcal{H}_i \setminus \{1\}$ and $B, E \in \mathcal{H}_{3-i}$ guaranteed by Lemma 2.9. We show that the element $A := \frac{C}{E}$ is the kind we are looking for. First observe

that by Proposition 2.1(i)

$$A = C \cdot \frac{1}{E} = B \cdot \frac{1}{C}$$

provide a $\mathcal{H}_i$ - and $\mathcal{H}_{3-i}$ -representation of A , respectively. Note furthermore that as $(\mathcal{H}_i \setminus \{1\}) \cap \mathcal{H}_{3-i} = \emptyset$, we have $C \neq E$ and hence $A \neq 1$. Consequently, by Proposition 2.1(ii) there exists a unique mixed representation $A = A_1 \cdot A_2$ with $A_i \in \mathcal{H}_i$.

Next we show that these six elements from the representations are all distinct.

Lemma 2.10. *The elements $A_i, C, \frac{1}{E} \in \mathcal{H}_i$ and $A_{3-i}, \frac{1}{C}, B \in \mathcal{H}_{3-i}$ are all distinct.*

To finish the proof of Theorem 2.3 note that by Proposition 2.4 these elements provide six distinct solutions of (6). $\square$

We finish this subsection proving the two lemmas from the above proof.

Proof of Lemma 2.10. For the distinctness first we establish that none of the six elements is 1. This is certainly true for C and $\frac{1}{C}$ by the choice of C in Lemma 2.9. Now assume that B or E is 1, say $B = 1$ (the argument in the case $E = 1$ is analogous). On the one hand, as $E \in \mathcal{H}_{3-i}$, by Proposition 2.7 E has a unique $\mathcal{H}_i$ -representation: $E = \frac{1}{E^q} \cdot \frac{1}{E^{q^2}}$. On the other hand $E = B \cdot E = C \cdot C$ is also a $\mathcal{H}_i$ -representation of E , so by the uniqueness we must have $E^q = \frac{1}{C} = E^{q^2} = \phi(E^q)$, and thus $E^q \in \mathbb{F}_q$ and $E^q = \frac{1}{C} \in \mathcal{H}_{3-i}$. That means $\phi(E) = E^q \in \mathbb{F}_q \cap \mathcal{H}_{3-i} = \mathcal{H}_1 \cap \mathcal{H}_2 = \{1\}$ by Proposition 2.1(iii), which is only possible if $E = 1$. This contradicts $C \neq 1$ and implies $B, E \neq 1$. Finally assume that A_1 or A_2 is equal to 1, say $A_1 = 1$. Then $A_1 \cdot A_2 = A = B \cdot \frac{1}{C}$ are two $\mathcal{H}_2$ -representations of A . By uniqueness either B or C should be 1, which is a contradiction by the above.

Since none of the six elements is 1 and by part (iii) of Proposition 2.1 $\mathcal{H}_1 \cap \mathcal{H}_2 = \{1\}$, we established that

$$\{A_i, C, \frac{1}{E}\} \cap \{A_{3-i}, B, \frac{1}{C}\} = \emptyset.$$

We are left to show that $|\{A_i, C, \frac{1}{E}\}| = 3$ and $|\{A_{3-i}, B, \frac{1}{C}\}| = 3$. Since they are proved analogously we present just the first one.

If $A_i = C$, then $A_{3-i} = \frac{1}{E}$, which is a contradiction as $A_{3-i} \in \mathcal{H}_{3-i}$ and $\frac{1}{E} \in \mathcal{H}_i$ and none of them is 1. If $A_i = \frac{1}{E}$, then $A_{3-i} = C$, which is a contradiction similarly as $A_{3-i} \in \mathcal{H}_{3-i}$ and $C \in \mathcal{H}_i$ and none of them is 1. Finally, suppose that $C = \frac{1}{E}$. Then we have $B = C^2 \cdot \frac{1}{E} = C^3$, which is a contradiction as $B \in \mathcal{H}_{3-i} \setminus \{1\}$ and $C^3 \in \mathcal{H}_i \setminus \{1\}$ because the polynomial h_i is defined over $\mathbb{F}_3$, hence if $h_i(C) = 0$, then $h_i(C^3) = 0$ as well. $\square$

Proof of Lemma 2.9. We want to find a $C \in \mathcal{H}_i \setminus \{1\}$ for $i = 1$ or 2 , such that C^2 has an $\mathcal{H}_{3-i}$ -representation with different elements B and E . This happens

exactly if one of the formulas in Proposition 2.6 is a nonzero square in $\mathbb{F}_{q^3}$ when we substitute $A = C^2$. It turns out that after simplifying the substituted formula of (i) using $h_2(C) = C^{q+1} + C^q + 1 = 0$ and clearing its square denominator we obtain the very same expression $D(C)$ as after simplifying the substituted formula of (ii) using $h_1(C) = C^{q+1} + C + 1 = 0$ and clearing its square denominator:

$$\begin{aligned} D = D(C) &= (C^2(C+1)^2 + (C+1)^2 - C^2)^2 - 4(C+1)^4 C^2 = \\ &= (C^2 + 3C + 1) \cdot (C^2 + C + 1) \cdot (C^2 + C - 1) \cdot (C^2 - C - 1). \end{aligned}$$

We aim to find an element $C \in \mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$ for which D is a square in $\mathbb{F}_{q^3}$, or equivalently, $N(D)$ is a square in $\mathbb{F}_q$.

As it turns out the factors of $N(D)$ can be conveniently expressed using the trace $Tr(C) = C + C^q + C^{q^2} =: t$ of C :

$$\begin{aligned} N(C^2 + 3C + 1) &= -t^2 - 3t - 1, \\ N(C^2 + C + 1) &= t^2 + 3t + 9, \\ N(C^2 + C - 1) &= t^2 + 3t + 1, \\ N(C^2 - C - 1) &= -t^2 - 3t - 1, \end{aligned}$$

and so

$$N(D) = (-t^2 - 3t - 1)^2(t^2 + 3t + 9)(t^2 + 3t + 1).$$

In characteristic 3 this expression is a square if and only if $t^2 + 1$ is a square. Using Theorem 5.18 from [20] we get that

$$\sum_{y \in \mathbb{F}_q} \eta(y^2 + 1) = -\eta(1) = -1,$$

where η is the quadratic character of $\mathbb{F}_q$. Therefore, as $y^2 + 1 = 0$ has at most two solutions, for at least $\frac{q-3}{2}$ elements $y \in \mathbb{F}_q$ the expression $y^2 + 1$ is a square.

This ensures the existence of many good “traces”, which we can use to construct many good C , as we now show that the trace function is a 3-to-1 function on $\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$. That is, if $Tr(C_1) = Tr(C_2)$ for $C_1, C_2 \in \mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$ then C_1 and C_2 are conjugates of each other. For this we note that the minimal polynomial of an element C of $\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$ can be expressed just by the trace t of C :

$$m_t(X) = (X - C)(X - C^q)(X - C^{q^2}) = X^3 - tX^2 - (t + 3)X - 1.$$

Hence if C_1 and C_2 have the same trace, then they have the same minimal polynomial.

Consequently there are exactly $\frac{|\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}|}{3} = \frac{2q}{3}$ elements in $\mathbb{F}_q^*$ that are traces of some element in $\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$.

In conclusion, there are at least $\left(\frac{2q}{3} + \frac{q-3}{2}\right) - q = \frac{q-9}{6} > 0$ elements $t \in F_q$ which are traces of an element C from $\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$, and for which $t^2 + 1$, and hence also $D = D(C)$, is a square. This completes the proof for $q > 9$.

Otherwise, by our assumption on q , we are left with the case $q = 9$. Then we can directly find a $t \in \mathbb{F}_9$ such that it is the trace of an element $C \in \mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$ and $t^2 + 1$ is a nonzero square in $\mathbb{F}_9$. In fact $t = -1$ will do. First note that $t^2 + 1 = 1 + 1 = 2$ is in $\mathbb{F}_3$ and hence is a square in its quadratic extension $\mathbb{F}_9$. Now, to finish the argument it is enough to show that $m_{-1}(X) = X^3 + X^2 + X - 1$ is the minimal polynomial of some $C \in \mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$ over $\mathbb{F}_9$, because then we automatically have $\text{Tr}(C) = t = -1$. It is immediate that $m_{-1}(X)$ is irreducible over $\mathbb{F}_3$, hence it can not have a root in $\mathbb{F}_9$, and therefore it is irreducible over $\mathbb{F}_9$. Next consider the polynomial $h_1(X)$ for $q = 3$. Then (when computed over $\mathbb{F}_3$) we have

$$h_1(X) = X^4 + X + 1 = (X - 1)(X^3 + X^2 + X - 1) = (X - 1)m_{-1}(X),$$

which in view of Proposition 2.2 means that the roots of $m_{-1}(X)$ solve the system (3) with norm function $N_{\mathbb{F}_{3^3}/\mathbb{F}_3}$. But then, as we have seen earlier, the roots of $m_{-1}(X)$ also solve the system (3) with norm function $N_{\mathbb{F}_{9^3}/\mathbb{F}_9}$, and hence, again by Proposition 2.2 and the fact $m_{-1}(1) \neq 0$, we have that any root of m_{-1} is in $\mathcal{H}_1 \cup \mathcal{H}_2 \setminus \{1\}$, as desired. $\square$

2.4 Norm equation systems and $K_{4,6}$ subgraphs in $\text{NG}(q, 4)$

In this subsection we prove Theorem 1.3. In the first lemma we connect solutions of a norm equation systems with three equations to four adjacencies in the projective norm-graph. For this let again $\mathbb{F}$ be an arbitrary field and $\mathbb{K}$ a Galois extension of degree 3.

Lemma 2.11. *Suppose that for $A_1, A_2, A_3 \in \mathbb{K}^*$ and $a_1, a_2, a_3 \in \mathbb{F}^*$, the element $Y \in \mathbb{K}^*$ is a solution of (5). Then in the projective norm graph $\text{NG}(\mathbb{F}, \mathbb{K})$ the vertex $\left(\frac{1}{Y}, \frac{1}{N(Y)}\right)$ is adjacent to all the vertices*

$$\left(\frac{1}{A_1}, \frac{a_1}{N(A_1)}\right), \quad \left(\frac{1}{A_2}, \frac{a_2}{N(A_2)}\right), \quad \left(\frac{1}{A_3}, \frac{a_3}{N(A_3)}\right), \quad (0, 1).$$

Proof. We check the adjacencies from the statement. For the first three vertices, using $N(Y + A_i) = a_i$, we have

$$N\left(\frac{1}{A_i} + \frac{1}{Y}\right) = N\left(\frac{Y + A_i}{A_i Y}\right) = \frac{N(Y + A_i)}{N(A_i Y)} = \frac{a_i}{N(A_i)} \cdot \frac{1}{N(Y)}.$$

For the last vertex we have

$$N\left(0 + \frac{1}{Y}\right) = N\left(\frac{1}{Y}\right) = 1 \cdot \frac{1}{N(Y)},$$

as requested. $\square$

The statement of Theorem 1.3 now follows easily.

Proof of Theorem 1.3. By Theorem 2.3 there exists an element $A \in \mathcal{N} \setminus \{1\}$ such that the system (6) admits 6 solutions in $\mathbb{F}_{q^3}$. We choose an element $C \neq 0, -1, -A$ from $\mathbb{F}_{q^3}$ such that $N(C) \neq 1$. This is clearly possible since $q^3 - 3 - (q^2 + q + 1) > 0$ for $q > 2$. Let $A_1 = C$, $A_2 = C + 1$, $A_3 = C + A$, $a_1 = 1$, $a_2 = -1$, and $a_3 = -1$. The corresponding norm equation system of the form (5), as the translation of (6) by C , also has six distinct solutions, which we denote by $Y_1, \dots, Y_6$. Since by the choice of C the elements A_1, A_2 , and A_3 , as well as the solutions $Y_1, \dots, Y_6$ are non-zero, Lemma 2.11 is applicable. As the elements A_1, A_2, A_3 are also distinct, so are the vertices

$$\left(\frac{1}{A_1}, \frac{a_1}{N(A_1)}\right), \quad \left(\frac{1}{A_2}, \frac{a_2}{N(A_2)}\right), \quad \left(\frac{1}{A_3}, \frac{a_3}{N(A_3)}\right), \quad (0, 1).$$

which we denote by (B_i, b_i) , $i = 1, 2, 3, 4$. By Lemma 2.11 each of them is adjacent to the six distinct vertices $\left(\frac{1}{Y_j}, \frac{1}{N(Y_j)}\right)$, $j = 1, \dots, 6$.

In order for these 24 adjacencies to indeed give rise to a $K_{4,6}$ in $NG(q, 4)$, we need to make sure that none of them represents a loop.

If $p = 2$, then $NG(q, 4)$ has no loop edges, so all the participating $4 + 6$ vertices are different, and they form a $K_{4,6}$.

Assume now that $p > 2$. Let us call an element $D \in \mathbb{F}_{q^3}$ *bad* for the pair of indices (i, j) , if $B_i + D = \frac{1}{Y_j} - D$. For every fixed pair (i, j) there exists exactly one D which is bad for (i, j) , namely $D = \frac{1}{2} \left(\frac{1}{Y_j} - B_i \right)$. Therefore, together there are at most 24 elements D which are bad for some pair (i, j) . As $q^3 \geq 27 > 24$, there exists an element $D \in \mathbb{F}_{q^3}$ which is not bad for any pair. For this D , the ten vertices $(B_i + D, b_i)$, $i = 1, 2, 3, 4$, and $\left(\frac{1}{Y_j} - D, \frac{1}{N(Y_j)}\right)$, $j = 1, \dots, 6$, are all distinct and hence give rise to a $K_{4,6}$. $\square$

3 General difference sets of Singer type

In general an (n, m, λ) difference set is a set $\mathcal{D}$ of m elements in a multiplicative group $\mathcal{G}$ of order n , such that any element $g \in \mathcal{G}$ has exactly λ mixed product representations with respect to $\mathcal{D}$. Singer's construction naturally generalizes to the case with parameters $(n, m, \lambda) = \left(\frac{q^t-1}{q-1}, \frac{q^{t-1}-1}{q-1}, \frac{q^{t-2}-1}{q-1}\right)$ for any $t \geq 3$, which are called Singer parameters. Unlike for $t = 3$, the planar case, for many values $t > 3$ difference sets having Singer parameters yet being inequivalent to Singer's construction are known to exist (see e.g. [13]).

In the case of planar difference sets we made use of a convenient description of the Singer difference set inside the multiplicative group $\mathbb{F}_{q^3}^* / \mathbb{F}_q^*$ as the collection of cosets of elements of trace 0. In what follows, we extend this description to the general setting of degree t cyclic Galois extensions over arbitrary fields $\mathbb{F}$, that we encountered already in Section 1 in connection with projective norm graphs. To this end, let $\mathbb{F}$ be an arbitrary field and $\mathbb{K}$ a cyclic Galois extension

of degree t with $t \geq 3$. Let $\text{Tr}_{\mathbb{K}/\mathbb{F}}$ be the trace function corresponding to this extension, i.e. for $A \in \mathbb{K}$ we have

$$\text{Tr}_{\mathbb{K}/\mathbb{F}}(A) = A + \phi(A) + \phi^{(2)}(A) + \cdots + \phi^{(t-1)}(A),$$

where $\phi^{(i)}$ denotes the i -fold iteration of some generator ϕ of the Galois group. Now consider the subset

$$\mathcal{S}_t = \{\overline{Y} \mid Y \in \mathbb{K}^*, \text{Tr}_{\mathbb{K}/\mathbb{F}}(Y) = 0\}$$

of the multiplicative group $\mathbb{K}^*/\mathbb{F}^*$, where $\overline{Y}$ denotes the image of $Y \in \mathbb{K}^*$ under the natural map $\mathbb{K}^* \rightarrow \mathbb{K}^*/\mathbb{F}^*$.

It is known [23] that when $\mathbb{F} = \mathbb{F}_q$ is a finite field then $\mathcal{S}_t$ is a Singer difference set with parameters $(\frac{q^t-1}{q-1}, \frac{q^{t-1}-1}{q-1}, \frac{q^{t-2}-1}{q-1})$. This can be extended for arbitrary $\mathbb{F}$, using the natural $(t-1)$ -dimensional projective space structure on $\mathbb{K}^*/\mathbb{F}^*$ (which is induced by the t -dimensional $\mathbb{F}$ -vector space structure of $\mathbb{K}$). It turns out that for any non-identity element $\overline{A}$, the set of $\mathcal{S}_t$ -elements from the mixed representations of $\overline{A}$ with respect to $\mathcal{S}_t$ forms a subspace of projective dimension $t-3$.

Proposition 3.1. *Let $\overline{\mathcal{L}}$ be a subspace of $\mathbb{K}^*/\mathbb{F}^*$ of projective dimension $t-2$. Then for every element $\overline{A} \in \mathbb{K}^*/\mathbb{F}^* \setminus \{\overline{1}\}$ the set*

$$\mathcal{R}_{\overline{\mathcal{L}}}(\overline{A}) = \{\overline{B} \in \overline{\mathcal{L}} \mid \exists \overline{C} \in \overline{\mathcal{L}} \text{ such that } \overline{A} = \overline{BC}^{-1}\} = \{\overline{B} \in \overline{\mathcal{L}} \mid \overline{B}/\overline{A} \in \overline{\mathcal{L}}\}$$

forms a subspace of projective dimension $t-3$.

Proof. Let $\mathcal{L}$ be a $(t-1)$ -dimensional subspace of $\mathbb{K}$ over $\mathbb{F}$ such that $\overline{\mathcal{L}} = \mathcal{L} \setminus \{0\} / \mathbb{F}^*$. Then for any element $\overline{A} \in \mathbb{K}^*/\mathbb{F}^* \setminus \{\overline{1}\}$ we have

$$\mathcal{R}_{\overline{\mathcal{L}}}(\overline{A}) = \mathcal{R}_{\mathcal{L}}(A) \setminus \{0\} / \mathbb{F}^*,$$

where $A \in \mathbb{K}^* \setminus \mathbb{F}^*$ is such that $\overline{A} = A \cdot \mathbb{F}^*$ and

$$\mathcal{R}_{\mathcal{L}}(A) = \{B \in \mathcal{L} \mid \exists C \in \mathcal{L} \text{ such that } A = BC^{-1}\}.$$

Observe that $\mathcal{R}_{\mathcal{L}}(A) = \mathcal{L} \cap A\mathcal{L}$, where $A\mathcal{L} = \{AL \mid L \in \mathcal{L}\}$.

Since $A \notin \mathbb{F}^*$, the $(t-1)$ -dimensional subspaces $\mathcal{L}$ and $A\mathcal{L}$ are different, hence their intersection has dimension $t-2$. Therefore the projective dimension of $\mathcal{R}_{\overline{\mathcal{L}}}(\overline{A})$ is indeed $t-3$. $\square$

Corollary 3.2. *Let $\overline{\mathcal{L}}$ be a subspace of $\mathbb{F}_{q^t}^*/\mathbb{F}^*$ of projective dimension $t-2$. Then $\overline{\mathcal{L}}$ is a difference set in the multiplicative group $\mathbb{F}_{q^t}^*/\mathbb{F}^*$ with parameters $(\frac{q^t-1}{q-1}, \frac{q^{t-1}-1}{q-1}, \frac{q^{t-2}-1}{q-1})$. In particular, so is $\mathcal{S}_t$.*

Proof. The set $\mathcal{R}_{\overline{\mathcal{L}}}(\overline{A})$ is in a one-to-one correspondence with the collection of mixed $\overline{\mathcal{L}}$ -representations of $\overline{A}$. Since a projective space of dimension $t-i$ over $\mathbb{F}_q$ has size $\frac{q^{t-i+1}-1}{q-1}$, the parameters of the difference set follow.

Finally note that as $\text{Tr}_{\mathbb{K}/\mathbb{F}} : \mathbb{K} \rightarrow \mathbb{F}$ is a non-trivial $\mathbb{F}$ -linear function, $\text{Ker}(\text{Tr}_{\mathbb{K}/\mathbb{F}})$ is a $(t - 1)$ -dimensional subspace of $\mathbb{K}$. Hence the set $\mathcal{S}_t = \text{Ker}(\text{Tr}_{\mathbb{K}/\mathbb{F}}) \setminus \{0\} / \mathbb{F}^*$ is a subspace of $\mathbb{K}^* / \mathbb{F}^*$ of projective dimension $t - 2$, and as such is a difference set with Singer parameters. $\square$

In Corollary 1.2 we gave an alternative description of planar difference sets of Singer type over finite fields as the root set of a simple polynomial and gave explicit formulas of the product representation of each element. Here we extend this result to the general setting. For this let $N_{\mathbb{K}/\mathbb{F}}$ denote the norm function of the extension $\mathbb{K}/\mathbb{F}$ and $\mathcal{N}$ the group of elements of norm 1 in $\mathbb{K}$. Furthermore, we shall consider the function

$$d_t(X) = 1 + X + X\phi(X) + X\phi(X)\phi^{(2)}(X) + \cdots + X\phi(X)\cdots\phi^{(t-2)}(X).$$

We remark that the function $d_t(X)$ appears in a paper of Foster [12] in the formulation of the Murphy condition.

In the next theorem we show that the set

$$\mathcal{D}_t = \{A \in \mathbb{K} \mid d_t(A) = 0\}$$

of roots of $d_t(Y)$ in $\mathbb{K}$ is contained in the multiplicative group $\mathcal{N}$ and has the same difference set property that the mixed representation of any element $A \in \mathcal{N} \setminus \{1\}$ with respect to $\mathcal{D}_t$ form (in some sense) a projective space of dimension $t - 3$ over $\mathbb{F}$. In addition we will also be able to describe concisely these product representations.

Theorem 3.3. *There is a group isomorphism $\bar{\Phi} : \mathbb{K}^* / \mathbb{F}^* \rightarrow \mathcal{N}$ such that $\bar{\Phi}(\mathcal{S}_t) = \mathcal{D}_t$. In particular, through $\bar{\Phi}$, the set $\mathcal{D}_t$ inherits the difference set property of $\mathcal{S}_t$ just like the projective space structure. Moreover, given an element $A \in \mathcal{N} \setminus \{1\}$ the different mixed representations of A with respect to $\mathcal{D}_t$ are exactly the products $B \cdot \left(\frac{B}{A}\right)^{-1}$, where B is a root in $\mathbb{K}$ of the function*

$$f_{t,A}(X) = d_t(X) - \frac{1}{\phi^{(t-1)}(A)} \cdot d_t\left(\frac{X}{A}\right).$$

Proof. Consider the $\mathbb{K}^* \rightarrow \mathbb{K}^*$ map Φ defined by $X \rightarrow \frac{\phi(X)}{X}$. On the one hand, one readily sees that the map Φ maps $\mathbb{K}^*$ into $\mathcal{N}$. On the other hand, by Hilbert's Theorem 90 [19] we know that for every $A \in \mathcal{N}$ there is an element $Y \in \mathbb{K}^*$ such that $A = \Phi(Y)$, which in turn shows that Φ is surjective. Therefore, as $\text{Ker}(\Phi) = \mathbb{F}^*$, the quotient map $\bar{\Phi} : \mathbb{K}^* / \mathbb{F}^* \rightarrow \mathcal{N}$ provides an isomorphism between the respective groups.

Next we show that the image of $\mathcal{S}_t$ under the map $\bar{\Phi}$ is $\mathcal{D}_t$. For this let $Y \in \mathbb{K}^*$. Then, on the one hand, we have

$$\begin{aligned} d_t(\Phi(Y)) &= d_t\left(\frac{\phi(Y)}{Y}\right) = 1 + \sum_{j=0}^{t-2} \prod_{i=0}^j \frac{\phi^{(i+1)}(Y)}{\phi^{(i)}(Y)} = \\ &= 1 + \sum_{j=0}^{t-2} \frac{\phi^{(j+1)}(Y)}{Y} = \frac{1}{Y} \text{Tr}_{\mathbb{K}/\mathbb{F}}(Y). \end{aligned}$$

Therefore, if $\overline{Y} \in \mathcal{S}_t$, then $\overline{\Phi}(\overline{Y}) \in \mathcal{D}_t$. Finally, let $A \in \mathcal{D}_t$, i.e. A is a root of d_t . Then $N_{\mathbb{K}/\mathbb{F}}(A) = 1 + A \cdot \phi(d_t(A)) - d_t(A) = 1$, and hence, again by Hilbert's Theorem 90, there is an element $Y \in \mathbb{K}^*$ such that $A = \Phi(Y)$ and so $A = \overline{\Phi}(\overline{Y})$. By the above calculations $\text{Tr}_{\mathbb{K}/\mathbb{F}}(Y) = Y \cdot d_t(\Phi(Y)) = Y \cdot d_t(A) = 0$, meaning that $\overline{Y} \in \mathcal{S}_t$. This concludes the proof of $\overline{\Phi}(\mathcal{S}_t) = \mathcal{D}_t$.

Now let us turn to the second part of the theorem. Given an element $A \in \mathcal{N} \setminus \{A\}$ first take a mixed representations $A = B \cdot C^{-1}$ with $B, C \in \mathcal{D}_t$. Then, in particular, we have $C = \frac{B}{A} \in \mathcal{D}_t$ and hence $d_t(B) = d_t\left(\frac{B}{A}\right) = 0$. Therefore, we have that B is also a root of the function

$$f_{t,A}(X) = d_t(X) - \frac{1}{\phi^{(t-1)}(A)} \cdot d_t\left(\frac{X}{A}\right).$$

For the other direction suppose that B is a root of $f_{t,A}$. Note that then necessarily $B \neq 0$, as otherwise we would have $0 = f_{t,A}(0) = 1 - \frac{1}{\phi^{(t-1)}(A)}$, which is a contradiction as for $A \in \mathcal{N} \setminus \{1\}$ we have $1 - \frac{1}{\phi^{(t-1)}(A)} \neq 0$. To finish the proof we need to show that $B, \frac{B}{A} \in \mathcal{D}_t$, as then the product $B \cdot \left(\frac{B}{A}\right)^{-1}$ is a valid product representation of A with respect to $\mathcal{D}_t$. Using that $N_{\mathbb{K}/\mathbb{F}}(A) = 1$ and $\phi^{(t)} \equiv \text{id}$, we have

$$\begin{aligned} 0 &= \phi(0) = \phi(f_{t,A}(B)) = \phi(d_t(B)) - \frac{1}{\phi^{(t)}(A)} \cdot \phi\left(d_t\left(\frac{B}{A}\right)\right) \\ &= \frac{1}{B} \cdot (d_t(B) + N_{\mathbb{K}/\mathbb{F}}(B) - 1) - \frac{1}{A} \cdot \frac{1}{\frac{B}{A}} \cdot \left(d_t\left(\frac{B}{A}\right) + N_{\mathbb{K}/\mathbb{F}}\left(\frac{B}{A}\right) - 1\right) \\ &= \frac{1}{B} \left(d_t(B) - d_t\left(\frac{B}{A}\right)\right) \implies d_t(B) = d_t\left(\frac{B}{A}\right), \end{aligned}$$

and hence $0 = f_{t,A}(B) = d_t(B) \left(1 - \frac{1}{\phi^{(t-1)}(A)}\right)$. However, as remarked earlier, for $A \in \mathcal{N} \setminus \{1\}$ we have $1 - \frac{1}{\phi^{(t-1)}(A)} \neq 0$, so this at once implies that $d_t(B) = d_t\left(\frac{B}{A}\right) = 0$, and so $B, \frac{B}{A} \in \mathcal{D}_t$, as required. $\square$

Next we spell out the special case of Theorem 3.3 when $\mathbb{F} = \mathbb{F}_q$, $\mathbb{K} = \mathbb{F}_{q^t}$ and ϕ is the Frobenius automorphism $X \rightarrow X^q$. This gives a description of the classic Singer difference set inside $\mathcal{N}$ as the set of roots of a simple polynomial and describes the mixed representations of any element also using the roots of a polynomial.

Corollary 3.4. *Let $q = p^k$ be a prime power, $t \geq 3$ an integer, and let us define over $\mathbb{F}_q$ the polynomial*

$$d_t(Y) = 1 + Y + Y^{1+q} + Y^{1+q+q^2} + \dots + Y^{1+q+\dots+q^{t-2}}$$

of degree $\frac{q^{t-1}-1}{q-1}$. Then the set

$$\mathcal{D}_t = \{A \in \mathbb{F}_{q^t} \mid d_t(A) = 0\}$$

of roots of $d_t(Y)$ forms a $(\frac{q^t-1}{q-1}, \frac{q^{t-1}-1}{q-1}, \frac{q^{t-2}-1}{q-1})$ -difference set in the cyclic group $\mathcal{N}$ of norm 1 elements of $\mathbb{F}_{q^t}$, which is equivalent to the Singer difference set $\mathcal{S}_t$. Moreover, given an element $A \in \mathcal{N} \setminus \{1\}$, the $\frac{q^{t-2}-1}{q-1}$ different mixed $\mathcal{D}_t$ -representations of A are exactly the products $B \cdot (\frac{B}{A})^{-1}$, where B is a root in $\mathbb{F}_{q^t}$ of the degree $\frac{q^{t-2}-1}{q-1}$ polynomial

$$f_{t,A}(X) = d_t(X) - A^{1+q+\dots+q^{t-2}} \cdot d_t\left(\frac{X}{A}\right)$$

□

In connection with Corollary 3.4 first note, that in particular it implies that the polynomials $d_t(X)$ and $f_{t,A}(X)$ always split over $\mathbb{F}_{q^t}$. Also, in the special case $t = 3$, we recover the difference set $\mathcal{H}_1$ from Corollary 1.2. In this case the polynomial $f_{t,A}$ is linear and its unique root is exactly the element A_1 from Corollary 1.2.

4 Concluding remarks

It is widely conjectured [5, 9] that the KST upper bound,

$$\text{ex}(n, K_{t,s}) \leq c_{t,s} n^{2-1/t}, \quad (9)$$

is tight up to constant factor for every $s \geq t \geq 2$. Together with the results of [2], Theorem 1.3 establishes that for $t = 4$, the projective norm graph $\text{NG}(q, 4)$ does not resolve this conjecture beyond the cases $s \geq 7$.

Several interesting questions remain open.

Complete bipartite graphs in projective norm graphs. Both in [2] and in Theorem 1.3 we could only find a special kind of copies of $K_{4,6}$. The number of these copies is only roughly q^7 . If that was it, then a simple uniform random subgraph of $\text{NG}(q, 4)$ with a few deleted edges would prove the tightness of the KST-bound for $t = 4$ and $s = 6$. We think however, also supported by computer experiments, that the number of copies of $K_{4,6}$ in $\text{NG}(q, 4)$ should be the same order as their typical number in the random graph of the same edge density.

Conjecture 1. *The number of copies of $K_{4,6}$ in $\text{NG}(q, 4)$ is $\Theta(q^{16})$.*

The determination of $s(t)$ is still widely open for $t \geq 5$, when we do not even know whether there is a $K_{t,t}$ in $\text{NG}(q, t)$ for every large enough q . While it is probably more realistic to expect that there are copies of $K_{t,(t-1)!}$ for every $t \geq 5$ and large enough q (besides numerology, i.e. that $s(t) = (t-1)!$ for $t = 2, 3$ and 4 , there are also algebro-geometric heuristics pointing towards this), we harbour a slim hope that $t = 4$ was still a special case. At least the graph $\text{NG}(q, 4)$ seems quite special, with a unique structure and symmetries, and maybe that alone is responsible for the presence of $K_{4,6}$ subgraphs.

Infinite projective norm graphs. The first constructions of dense $K_{t,t}$ -free graphs were motivated by simple facts from real Euclidean geometry: two lines of the plane intersect in at most one point; three unit spheres in 3-space intersect in at most two points. Consequently the point/line incidence graph of the Euclidean plane is $K_{2,2}$ -free, and the unit-distance graph of the Euclidean 3-space is $K_{3,3}$ -free. Furthermore these infinite $K_{t,t}$ -free graphs are “dense” in terms of the dimension of the neighborhoods. So when defined over appropriate finite fields, in a way that the algebra in the proof of their $K_{t,t}$ -freeness carries over, their number of edges verifies the tightness of the KST-bound for $t = 2$ and $t = 3$.

The (projective) norm graphs were not constructed this way, yet one can define them over an arbitrary field $\mathbb{F}$, see Section 1. The key lemma from [17] holds over any field, which implies that the proof of the $K_{t,(t-1)!+1}$ -freeness of $\text{NG}(q, t)$ also extends to $\text{NG}(\mathbb{F}, \mathbb{K})$ for arbitrary $\mathbb{F}$ and arbitrary Galois extension $\mathbb{K}$ of degree $t - 1$. (For completeness we include the argument in the Appendix.) In particular, if $t = 4$ then $\text{NG}(\mathbb{F}, \mathbb{K})$ does not contain $K_{4,7}$ for any field $\mathbb{F}$. After seeing that $\text{NG}(q, 4)$ does contain $K_{4,6}$ for any $q > 4$, it might seem plausible that the same is true for infinite fields.

The tightness of the KST-bound. The tightness of the order of magnitude of the KST-bound is a central question of the area. This conjecture suggests that whatever density is not ruled out by simple double counting, should essentially be possible to realize with a construction. Here we speculate that this might not be the case and offer a counter-conjecture.

In any graph with $cn^{7/4}$ edges, the number of common neighbors of an average 4-tuple is (at least) a constant c' depending on c . If this graph with $cn^{7/4}$ edges is random then this constant average is spread out over $\binom{n}{4}$ distributions that are each approximately Poisson with mean c' . Consequently for any s , a positive constant proportion of 4-tuples have at least s neighbors. In contrast, in any $K_{4,s}$ -free construction with $cn^{7/4}$ edges (matching the KST-bound), no 4-tuple can have more than $s - 1$ neighbors. So in such constructions each of the Poisson-tails has to be absorbed by the 4-tuples with at most $s - 1$ common neighbors. Should such graphs exist for some s , they must be extremely rare, their mere existence has to be a coincidence and should require quite a bit of structure.

In all known constructions (including Klein [10], Brown [7], KRS [17], ARS [1] and Bukh [8]) this is realized using the algebro/geometric notion of dimension and its strong correlation with the cardinality of the corresponding variety: an “everyday” d -dimensional variety over $\mathbb{F}_q$ has roughly $\Theta(q^d)$ points. To achieve that the common neighborhood of four vertices is less than a constant s , one appeals to the geometric intuition that in the four-dimensional space the intersection of four hypersurfaces, that are in general enough position, is 0-dimensional, and hence it is the union of constantly many points. A graph can be defined on a four-dimensional space of roughly $q^4 =: n$ vertices, and the neighborhood of each vertex can be chosen to be some hypersurface, which then have

roughly the desired size $q^3 = n^{3/4}$. For a $K_{4,s}$ -free graph the intersection of any four of the neighborhood-hypersurfaces should have size $< s$. Now if the neighborhood-hypersurfaces are carefully chosen, so that any four of them are in general enough position, then their intersection is 0-dimensional and hence has size $\Theta(q^0)$, a constant.

How to choose the hypersurfaces and what is this constant? Even though choosing randomly is a generally good strategy (witnessed by the random algebraic construction of Bukh [8]), finding good explicit choices, as it is often the case, is not so straightforward. By the KST-bound the constant bounding the neighborhoods of t -tuples in any graph with $cn^{2-1/t}$ edges is at least $t - 1$, and the projective norm graph chooses neighborhoods where they are bounded by not more than $(t - 1)!$. The current analysis of the random choice gives an upper bound of $t^{O(t)}$.

Now how small could this constant be? We believe that the presence of some notion of “dimension” in this problem is a necessity and this constant is just going to be in the nature of the geometry of the hypersurface-neighborhoods we have chosen. As such, it will not just be limited by the simple combinatorial restrictions of the KST-bound but also by those of geometry/algebra. And then its extrema should be delivered by a regular, rigid structure with distinctive properties. For $t = 4$ we have seen ample evidence that the projective norm graph $\text{NG}(q, 4)$ fits this bill, and tend to accept it as the limit of what algebra can offer in this realm. Since we know now that $K_{4,6}$ does occur in $\text{NG}(q, 4)$, we conjecture the following.

Conjecture 2. $\text{ex}(n, K_{4,6}) = o(n^{7/4})$.

We note that should this conjecture be true, it of course implies that the KST-bound is not tight for the symmetric case $K_{4,4}$ either. That further implies that $\text{ex}(n, K_{t,t}) = o(n^{2-1/t})$ for every $t \geq 5$; this is the consequence of (an adaptation of) a theorem of Erdős and Simonovits [11].

While we do believe Conjecture 2, at the same time we also think that it is more likely that we see it disproved than proved. For a proof one might need to develop a two step approach. Given a $K_{4,4}$ -free graph with $cn^{7/4}$ edges, build up a significant-enough proportion of a pseudo-algebraic/geometric framework using the neighborhoods as hypersurfaces, with surfaces having appropriate intersection sizes and structure. Then, provided the pseudo-algebra/geometry gives a structure rigid enough, establish the existence of a $K_{4,4}$. Preliminary results in this direction were proven by Blagojevic, Bukh, and Karasev [4] and in this paper.

References

- [1] N. Alon, L. Rónyai, T. Szabó. Norm-graphs: variations and applications. *Journal of Combinatorial Theory, Series B* **76**:280–290 (1999).
- [2] T. Bayer, T. Mészáros, L. Rónyai, T. Szabó. Exploring projective norm graphs. *arXiv:1908.05190* (2019)

- [3] G. Berman. Finite projective plane geometries and difference sets. *Transactions of the American Mathematical Society* **74**:492–499 (1953).
- [4] P. Blagojević, B. Bukh, R. Karasev. Turán numbers for $K_{s,t}$ -free graphs: topological obstructions and algebraic constructions. *Israel Journal of Mathematics* **197**(1):199–214 (2013).
- [5] B. Bollobás. *Extremal Graph Theory*. Academic Press (1978).
- [6] W. Bosma, J. Cannon, C. Playoust. The Magma algebra system I -The user language. *Journal of Symbolic Computation* **24**:235–265 (1997).
- [7] W.G. Brown. On graphs that do not contain a Thomsen graph. *Canadian Mathematical Bulletin* **9**:281–285 (1966).
- [8] B. Bukh. Random algebraic construction of extremal graphs. *Bullettin of the London Mathematical Society* **47**(6):939–945 (2015).
- [9] F.R.K. Chung, R. Graham. *Erdős on Graphs - His Legacy of Unsolved Problems*. A.K. Peters, Wellesley (1998).
- [10] P. Erdős. On sequences of integers no one of which divides the product of two others and related problems. *Mitt. Forsch. Institut Mat. und Mech. Tomsk* **2**:74–82 (1938).
- [11] P. Erdős, M. Simonovits. A limit theorem in graph theory. *Studia Scientiarum Mathematicarum Hungarica* **1**:215–235 (1966).
- [12] L. Foster. HT90 and “simples” number fields. *Illinois Journal of Mathematics* **55**(4):1621–1655 (2011).
- [13] B. Gordon, W.H. Mills, L.R. Welch. Some new difference sets. *Canadian Journal of Mathematics* **14**:614–625 (1962).
- [14] C. Grosu. A note on projective norm graphs. *International Journal of Number Theory* **14**(1):55–62 (2018).
- [15] H. Halberstam, R.R. Laxton. Perfect difference sets. *Glasgow Mathematical Journal* **6**(4):177–184 (1964).
- [16] D.R. Hughes. A note on difference sets. *Proceedings of the American Mathematical Society* **6**(5):689–692 (1955).
- [17] J. Kollár, L. Rónyai, T. Szabó. Norm-graphs and bipartite Turán numbers. *Combinatorica* **16**:399–406 (1996).
- [18] T. Kővári, V.T. Sós, P. Turán. On a problem of K. Zarankiewicz. *Colloquium Mathematicae* **3**(1):55–57 (1954).
- [19] S. Lang. *Algebra*. Springer-Verlag, New York (2002).

- [20] R. Lidl, H. Niederreiter. Introduction to finite fields and their applications. Cambridge University Press (1986).
- [21] E.H. Moore, H.S. Pollatsek. Difference sets: Connecting Algebra, Combinatorics, and Geometry. Student Mathematical Library **67**, American Mathematical Society (2013).
- [22] T. Nagell. Sur un type particulier d'unités algébriques. Arkiv för Matematik **8**(2):163–184 (1970).
- [23] A. Pott. Finite Geometry and Character Theory. Lecture Notes in Mathematics **1601**, Springer-Verlag, Berlin, Heidelberg (1995).
- [24] D. Shanks. The Simplest Cubic Fields. Mathematics of Computation **28**(128):1137–1152 (1974).
- [25] J. Singer. A Theorem in Finite Projective Geometry and Some Applications to Number Theory. Transactions of the America Mathematical Society **43**(3):377–385 (1938).

5 Appendix

5.1 $K_{4,7}$ -freeness of $\text{NG}(\mathbb{F}, \mathbb{K})$

As before, let $\mathbb{F}$ be an arbitrary field and for $t \geq 2$ let $\mathbb{K}$ be a cyclic Galois extension of degree $t - 1$, whose Galois group is generated by the automorphism ϕ . We aim to prove that the projective norm graph $\text{NG}(\mathbb{F}, \mathbb{K})$ is $K_{t, (t-1)!+1}$ -free, i.e. that the common neighbourhood of t vertices has size at most $(t - 1)!$. The proof is exactly the same as in the case of finite fields, and it is included here only because we need some of the steps anyway.

For $\ell \geq 2$ let $U = \{(B_i, b_i) \mid 1 \leq i \leq \ell\}$ be an ℓ -set of vertices. Without loss of generality we may assume that $B_i \neq B_j$ for $i \neq j$, as otherwise the common neighbourhood of U would be empty. For $1 \leq i \leq \ell - 1$ let

$$A_i = A_i(U) := \frac{1}{B_i - B_\ell} \in \mathbb{K} \quad \text{and} \quad a_i = a_i(U) := \frac{b_i}{b_\ell} \cdot \text{N}(B_i) \in \mathbb{F}^*,$$

and consider the system

$$\text{N}(X + A_i) = a_i \quad 1 \leq i \leq \ell - 1 \tag{10}$$

of norm equations. Simple substitutions show that a vertex (X, x) is in the common neighbourhood of U if and only if $\xi((X, x)) := \frac{1}{X + B_\ell}$ is a solution to (10). Note that $\xi((X, x))$ is well-defined, as $\text{N}(X + B_\ell) = b_\ell \cdot x \neq 0$. Therefore, we have that the size of the common neighbourhood of U is at most the number of solutions to (10). We remark that it can be less (by one) exactly if 0 is a solution to (10), which happens if $a_1 = a_2 = \dots = a_\ell$.

Now set $\ell = t$ and let us recall the key lemma from the original proof due to Kollár, Rónyai and Szabó [17].

Lemma 5.1. *Let $\mathbb{K}$ be a field and $\alpha_{i,j}, \beta_j \in \mathbb{K}$ for $1 \leq i, j \leq m$ such that $\alpha_{i_1,j} \neq \alpha_{i_2,j}$ for $i_1 \neq i_2$. Then the system of equations*

$$(x_1 - \alpha_{i,1})(x_2 - \alpha_{i,2}) \cdots (x_m - \alpha_{i,m}) = \beta_i \quad i = 1, 2, \dots, m$$

has at most $m!$ solutions $(x_1, x_2, \dots, x_m) \in \mathbb{K}^m$.

To finish the proof of the $K_{t,(t-1)!+1}$ -freeness we just need to realize that the equations in (10) can be rewritten, namely for $1 \leq i \leq t-1$ we have

$$\begin{aligned} N(X + A_i) &= (X + A_i) \cdot \phi(X + A_i) \cdots \phi^{(t-2)}(X + A_i) = \\ &= (X + A_i) (\phi(X) + \phi(A_i)) \cdots (\phi^{(t-2)}(X) + \phi^{(t-2)}(A_i)) = a_i. \end{aligned}$$

Now if for $1 \leq i, j \leq t-1$ we set $\alpha_{i,j} = \phi^{(j-1)}(A_i)$, $x_j = \phi^{(j-1)}(X)$ and $\beta_i = a_i$ then Lemma 5.1 applies with $m = t-1$ and gives that the system (10) has at most $(t-1)!$ solutions and hence any t -set of vertices in $\text{NG}(\mathbb{F}, \mathbb{K})$ has at most $(t-1)!$ common neighbours, as desired.